

Session 5 · Agents, Governance, Your Capstone

Until now you wrote every plan. Now the model plans its own steps, within limits you define.

Plan: concepts 10' · demo 18' · lab 22' · capstones 20' · close 5'

By the end of Session 5 you can

1. **Explain** what separates an agent from a workflow, and when each wins
2. **Read** an agent loop and point at every governance lever in the code
3. **Run** a governed agent on your own company pair, and pass its human gate
4. **Present** your AI workflow in five minutes, with a trust story

Workflow vs agent

	Workflow (Session 4)	Agent (now)
The plan	fixed, written by you	chosen by the model , step by step
Tools	your code calls them	the model requests, your code executes
Stops when	the script ends	when it decides, or at the limits you set
Use for	repeatable work	paths that depend on the data

The agent mechanism is a loop

```
while steps < MAX_STEPS:
    response = claude(messages, tools=TOOLS)
    if response.wants_tool:
        result = run_tool(...) # YOUR code executes
        messages += [response, result] # the conversation IS the memory
    else:
        break
```

About forty lines. Once you have read them, you can evaluate any vendor's agent claims from first principles — the notebook then shows the same discipline in **PydanticAI** and maps every lever to **LangGraph**.

Governance: six levers, all in code

Lever	Failure it prevents
MAX_STEPS	the agent looping indefinitely
TOKEN_BUDGET	unbounded spend
Tool whitelist	doing anything you gave no tool for
Forced output schema	an essay instead of a decision
Human gate	anything saved without your yes
Audit trail	"why did it do that?" with no answer

The demonstration: a currency trap

Task: compare Novo Nordisk with Eli Lilly.

naive agent:	DKK 300bn vs USD 65bn → "Novo is 5x bigger"	(wrong)
our agent:	tool detects mixed units → warning → compares growth and margins only	(correct)

- **Defense in depth:** a rule in the prompt AND a check in the tool
- You build both protections yourself, then run your own pair

The libraries in this session

Library	Role here	Standing
<code>pydantic</code>	the typed recommendation	the industry standard for validation
PydanticAI	the agent loop, packaged (live demo)	state-of-the-art agent framework, by the Pydantic team
LangGraph	graph agents (mapped, not built)	the other leading agent framework

How the labs work

Every exercise sits between two markers. **You fill the gaps. Nothing else changes.**

```
### START CODE HERE ###  
"3. ALWAYS check the [WHICH FIELD?] field before comparing figures."  
### END CODE HERE ###
```

- **None** → replace with the correct column, value or variable
- **[QUESTION IN CAPITALS]** → replace with the text the bracket asks for
- **Everything else is given.** Do not rewrite it.
- Then run the  **check cell** directly below. Green means correct: continue.

Stuck for two minutes? Select the lines, press **Option+K** (**Alt+K** on Windows), and ask the ***** panel.

Your lab, then your five minutes

Lab (22'): build the currency tool + the agent's rules → run YOUR pair → pass the gate

Capstone (5', hard stop):

1. The **job** your workflow does
2. The **live run**
3. The **trust story**: one failure you caught + the check that caught it
4. One honest **number**
5. The **next step**

The trust story carries the most weight in grading. A documented, caught failure is stronger evidence than an unexamined success.

What you own now

Grounded prompts · a comparable-company analysis tool · a tested discounted-cash-flow model · an earnings engine with fabrication detection · a screener on live regulatory filings · a governed agent

- **A challenge for the coming week:** choose one recurring task at your desk, build it as a workflow, and show a colleague

Never ship a number you haven't verified.